

Taylor Series Worksheet

Twenty problems on known series, transformations, coefficients, intervals, approximations, and error.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/taylor-series/>

Name: _____ **Date:** _____

1. Find the degree-3 Maclaurin polynomial for e^x .

2. Find the degree-5 Maclaurin polynomial for $\sin x$.

3. Find the degree-4 Maclaurin polynomial for $\cos x$.

4. Write the first four nonzero terms of $\ln(1+x)$.

5. Expand $1/(1-x)$ as a power series.

6. Expand $1/(1+x^2)$ as a power series.

7. Expand $1/(1-3x)$ as a power series.

8. Find a power series for $1/(1-x)$ to the power 2.
9. Find a power series for $\arctan x$ by integrating $1/(1+x^2)$.
10. Find the degree-2 Taylor polynomial for $\ln x$ centered at 1.
11. Write the Taylor series for e^x centered at a .
12. If $f^{(n)}(0) = 2^n$, find the Maclaurin series.
13. Use T_3 for e^x to approximate $e^{0.1}$.
14. Use $x - x^3/6$ to approximate $\sin(0.2)$.
15. Find the radius of $\sum nx^n/4^n$.
16. Find the interval of convergence of $\sum x^n/n$.

17. Find the interval of convergence of $\sum (x - 2)^n / (n3^n)$.
18. Find R for $\sum n!(x + 1)^n / (2n)!$.
19. Bound the error of the degree-3 Maclaurin approximation to e to the power (0.2).
20. How many terms of the alternating arctan series ensure error below 0.001 at $x=1/2$?